

MINHAZUL ABEDIN

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EDUCATION

Bachelor of Science in Computer Science and Telecommunication Engineering

Jan 2017 – March 2022

Noakhali Science and Technology University

Noakhali, Bangladesh

- CGPA: 3.58/4.0 (**3.98** in last 2 semesters)
- Merit Position: 6th of 54 (Top 11% of cohort)

RESEARCH EXPERIENCE

Research Collaborator

December 2020 – September 2021

Darul Irfan Research Institute

Chattogram, Bangladesh

- Collaborated with institute members for research upskilling, and co-authored a peer-reviewed journal publication under the guidance of **Dr. Asif Mahbub Karim**.

TEACHING EXPERIENCE

- **Mentor:** Mentored software engineering interns through an industry-oriented curriculum, guiding them in API design, database modeling, and production-grade system architecture. Conducted code reviews, performance evaluations, and provided technical feedback.
- **Instructor:** Taught Physics, Chemistry, and Mathematics to secondary and higher secondary students, helping them develop strong analytical and problem-solving skills.

RESEARCH WORK & PUBLICATIONS

Publications

- Hoque, S. I., Abedin, M. M., & Chowdhury, M. S. (2021). **Delving into Blood Transfusions Data Through Data Mining: A Study of Maizbhandari Shah Emdadia Blood Donors Group to Select Volunteer Blood Donors Efficiently.** *American International Journal of Multidisciplinary Scientific Research*, 10(1), 1–11.
<https://doi.org/10.46281/aijmsr.v10i1.1308>

Undergraduate Thesis

- **A Comparative Study of Lightweight Face Mask Detectors for Real-Time Applications.** [\[pdf\]](#)
Supervisor: Abul Kalam Azad (Assistant Professor, Dept. of CSTE, NSTU; Ph.D. Candidate in Computer Science, The University of Alabama)

Writing Samples

- **Investigating Manual and Automatic Feature Extraction Models in Evaluating Breast Invasive Ductal Carcinoma Grading.** [\[pdf\]](#)

Presentations

- **Overlapping Vehicles Detection Using Different Object Detection Methodologies.** (*Presented in Big Data and Machine Learning (BDML 2020) workshop*) [\[link\]](#)

PROFESSIONAL EXPERIENCE

Software Engineer

Jan 2023 – Present

AIT Inc (www.ait.inc)

Dhaka, Bangladesh

- **Worked over 3.5 years as a Backend Developer**, building and maintaining systems across diverse business domains — including sales, CRM, file and access management, e-commerce, learning management systems, and budget forecasting — for national and international clients, including the USA and Australia.
- Led integration of LLM-based automation into a budget forecasting platform, cutting manual data preparation time by over 90%.
- Integrated AI-assisted tooling into the development workflow — including code review, ticket creation and resolution, documentation, technical diagrams, and database design — reducing engineering overhead by 2–4 hours daily.
- Mentored junior developers, conducted code reviews and technical interviews, and collaborated with cross-functional teams across QA, UI/UX, AI, Mobile, and Business to support team growth and project delivery.

TECHNICAL SKILLS

- **Languages:** Python | C#
- **AI/ML Frameworks & Tools:** PyTorch | TensorFlow/Keras | Scikit-Learn | Pandas | NumPy | OpenCV | Hugging Face | OpenAI API | Ollama | RoboFlow
- **Data & Cloud:** PostgreSQL + pgvector (vector embeddings) | MongoDB | Redis | Cassandra | AWS
- **Web Frameworks:** .NET (Core, MVC) | React.js, Next.js

PROJECTS

- **Automated Grading of Breast Invasive Ductal Carcinoma:** Built a classification pipeline using Fisher Vector encoding and ResNet-18 for histopathology grading. [\[github\]](#)
- **Discord Bot for AI Search & QA:** Built a bot for image similarity search, text summarization, and question answering using Ollama and pgvector for vector-based retrieval. [\[github\]](#)
- **Custom Object Detection:** Implemented YOLO and SSD object detection architectures from scratch for face mask detection. [\[github\]](#)
- **Traffic Pattern Analysis:** Trained and evaluated object detection models to analyze traffic patterns in Bangladesh's first international AI traffic detection competition. [\[github\]](#)
- **Credit Approval Prediction:** Built supervised and semi-supervised models (Logistic Regression, Decision Trees, SVM, fuzzy classifiers) for credit approval prediction.

CERTIFICATIONS

- **AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents** — Udemy, 2025
- **Deep Learning Specialization** — Coursera, 2020
- **Applied Machine Learning in Python** — Coursera, 2020

HONORS & AWARDS

- **Best Project Award, Audit Week** — Astha IT, 2025
- **Department Scholarship for Outstanding Academic Performance** (awarded 4 times) — 2017–2022
- **Board Merit Scholarship, Secondary School** — 2011
- **Intra-departmental Programming Contest, 3rd Place** — 2018

OTHER PUBLICATIONS

- Jahan, S. N., Abedin, M. M., & Mizan, A. K. (2022). **Attitudes and Involvements in Maizbhandari Sufi Music of the People of Chattogram: A Critical Analysis.** *Darul Irfan Research Journal*, 2(1). <https://doi.org/10.66568/dirjv2i1a04>

EXTRA-CURRICULAR ACTIVITIES

Member, Organisation of Engineers of Bangladesh **2017 – Present**

- Collaborated with engineers from diverse disciplines on community welfare initiatives, including merit-based financial scholarships for underprivileged students, COVID-19 relief through PPE distribution, and sustained social outreach programs supporting vulnerable communities.

Member, Dhaka Ghan Bhandar Phataghar **2025 – Present**

- Volunteering with a cross-functional team of students and professionals to empower the local community through academic tutoring, technical skill workshops, and structured career mentorship for students and guardians — with a focus on cultivating industry-ready graduates.